

# Single-GPU Multilingual Speech (STT/TTS) Concurrent Stress, Load, Spike & Volume Benchmark Report

**Date:** 2026-08-21  
**Isolated Target GPU:** NVIDIA RTX A6000 (GPU 2, `CUDA_VISIBLE_DEVICES="2"`)  
**Languages Evaluated:** English ( `en` ), Luganda ( `lg` ), Swahili ( `sw` )  
**Scope:** Per-Language Concurrent Curves ( $\$c=10$  to 100\$), Mixed Multilingual Stress ( $\$c=10$  to 250\$), Instantaneous Spike Burst ( $\$c=250$ \$), High-Volume Soak ( $\$n=1,500$  requests), User vs Staff Isolation, and Scoped GPU Cleanup ( `~/Mpairwe7` only).

## 1. Executive Summary

This report documents the performance, capacity boundaries, and concurrency envelopes of the URA Taxpayer Chatbot's multilingual speech layer on a **single isolated NVIDIA RTX A6000 GPU (GPU 2)**. The system was benchmarked across simultaneous Speech-to-Text (ASR) and Text-to-Speech (TTS) pipelines for **English, Luganda, and Swahili**, evaluating per-language scaling, mixed multilingual traffic spikes, long-duration soak, and tenant isolation between citizen voice sessions and staff operations.

### Key Highlights

- Per-Language Throughput:** Sustained **94.0 – 144.2 RPS** per language with character generation rates reaching **6,075 – 9,632 chars/sec**.
- Multilingual Mix Concurrency ( $\$c = 250$ \$):** Sustained **101.5 RPS** under heavy mixed traffic across all three languages with **0.0% error rate**.
- Instantaneous Spike Burst ( $\$c = 250$ \$ in 50ms):** Processed 250 concurrent speech requests in **2.38s (105.0 RPS)** with zero dropped connections and zero circuit breaker trips.
- Sustained Soak (1,500 Requests):** Maintained continuous multilingual speech processing with **0.0 MB memory leak**.
- User vs Staff Isolation:** 600 mixed requests (300 User voice + 300 Staff admin) executed concurrently in **2.18s** with **0 cross-tenant violations** and  $<190\text{ms}$  median latency.
- Scoped GPU Cleanup:** Hardware verified cleanly deallocated post-test strictly for `~/Mpairwe7` workspace processes with zero interference to external system services.

## 2. Per-Language Concurrent Load Curves (English, Luganda, Swahili)

### 2.1 English ( `en` ) — URA Invoicing & PAYE Tax Queries

| Concurrency (\$c\$) | Requests | Duration (s) | Throughput (RPS) | Char Throughput | Median (p50) | Latency p90 | Latency p95 | Latency p99 | Error Rate |
|---------------------|----------|--------------|------------------|-----------------|--------------|-------------|-------------|-------------|------------|
| $\$c = 10$ \$       | 40       | 0.320s       | 125.1            | 8,083 char/s    | 69.5 ms      | 107.6 ms    | 109.3 ms    | 110.1 ms    | 0.0%<br>✓  |
| $\$c = 25$ \$       | 50       | 0.532s       | 94.0             | 6,075 char/s    | 282.0 ms     | 330.5 ms    | 334.1 ms    | 339.8 ms    | 0.0%<br>✓  |
| $\$c = 50$ \$       | 100      | 0.874s       | 114.4            | 7,390 char/s    | 403.3 ms     | 428.0 ms    | 431.7 ms    | 448.9 ms    | 0.0%<br>✓  |

| Concurrency (\$c\$) | Requests | Duration (s) | Throughput (RPS) | Char Throughput | Median (p50) | Latency p90 | Latency p95 | Latency p99 | Error Rate        |
|---------------------|----------|--------------|------------------|-----------------|--------------|-------------|-------------|-------------|-------------------|
| \$c = 100\$         | 200      | 1.964s       | 101.8            | 6,580 char/s    | 847.0 ms     | 1,103.8 ms  | 1,112.8 ms  | 1,124.9 ms  | 0.0% <div>✓</div> |

2.2 Luganda ( lg ) — Native Ugandan Luganda Tax Queries (Omusolo)

| Concurrency (\$c\$) | Requests | Duration (s) | Throughput (RPS) | Char Throughput | Median (p50) | Latency p90 | Latency p95 | Latency p99 | Error Rate        |
|---------------------|----------|--------------|------------------|-----------------|--------------|-------------|-------------|-------------|-------------------|
| \$c = 10\$          | 40       | 0.408s       | 98.0             | 6,309 char/s    | 72.0 ms      | 201.7 ms    | 206.9 ms    | 207.3 ms    | 0.0% <div>✓</div> |
| \$c = 25\$          | 50       | 0.384s       | 130.2            | 8,382 char/s    | 177.3 ms     | 195.4 ms    | 196.7 ms    | 203.8 ms    | 0.0% <div>✓</div> |
| \$c = 50\$          | 100      | 0.755s       | 132.4            | 8,525 char/s    | 352.3 ms     | 382.6 ms    | 384.5 ms    | 387.3 ms    | 0.0% <div>✓</div> |
| \$c = 100\$         | 200      | 1.810s       | 110.5            | 7,114 char/s    | 796.3 ms     | 882.7 ms    | 922.2 ms    | 936.7 ms    | 0.0% <div>✓</div> |

2.3 Swahili ( sw ) — Regional East African Kiswahili (Kodi)

| Concurrency (\$c\$) | Requests | Duration (s) | Throughput (RPS) | Char Throughput | Median (p50) | Latency p90 | Latency p95 | Latency p99 | Error Rate        |
|---------------------|----------|--------------|------------------|-----------------|--------------|-------------|-------------|-------------|-------------------|
| \$c = 10\$          | 40       | 0.277s       | 144.2            | 9,632 char/s    | 63.0 ms      | 81.9 ms     | 87.5 ms     | 91.6 ms     | 0.0% <div>✓</div> |
| \$c = 25\$          | 50       | 0.376s       | 133.1            | 8,894 char/s    | 169.8 ms     | 190.0 ms    | 194.7 ms    | 198.7 ms    | 0.0% <div>✓</div> |
| \$c = 50\$          | 100      | 0.935s       | 107.0            | 7,146 char/s    | 535.8 ms     | 559.4 ms    | 562.8 ms    | 568.0 ms    | 0.0% <div>✓</div> |
| \$c = 100\$         | 200      | 1.962s       | 101.9            | 6,809 char/s    | 896.4 ms     | 1,062.5 ms  | 1,087.0 ms  | 1,117.0 ms  | 0.0% <div>✓</div> |

3. Concurrent Multilingual Mix Load & Stress Matrix (EN + LG + SW)

Simultaneous mixed speech requests distributed equally across English, Luganda, and Swahili:

| Mixed Concurrency (\$c\$) | Total Requests | Duration (s) | Throughput (RPS) | Median Latency (p50) | Latency p90 | Latency p95 | Latency p99 | Error Rate        |
|---------------------------|----------------|--------------|------------------|----------------------|-------------|-------------|-------------|-------------------|
| \$c = 10\$                | 50             | 0.371s       | 134.8 RPS        | 70.7 ms              | 87.7 ms     | 93.6 ms     | 101.0 ms    | 0.0% <div>✓</div> |
| \$c = 50\$                | 100            | 0.982s       | 101.8 RPS        | 497.0 ms             | 582.9 ms    | 593.2 ms    | 596.9 ms    | 0.0% <div>✓</div> |

| Mixed Concurrency (\$c\$) | Total Requests | Duration (s) | Throughput (RPS) | Median Latency (p50) | Latency p90 | Latency p95 | Latency p99 | Error Rate |
|---------------------------|----------------|--------------|------------------|----------------------|-------------|-------------|-------------|------------|
| \$c = 100\$               | 200            | 1.672s       | 119.6 RPS        | 848.1 ms             | 931.4 ms    | 942.6 ms    | 951.3 ms    | 0.0%<br>✓  |
| \$c = 250\$               | 500            | 4.928s       | 101.5 RPS        | 2,159.2 ms           | 2,824.2 ms  | 2,965.1 ms  | 3,147.7 ms  | 0.0%<br>✓  |

#### 4. Instantaneous Traffic Spike Surge & High-Volume Soak

- **Traffic Spike Burst (\$c = 250\$ in 50ms):** Processed 250 simultaneous multilingual requests in **2.380s (105.0 RPS, p50: 1,874.2ms, p95: 1,956.3ms, 0% errors)**. Zero circuit breaker trips.
- **Sustained High-Volume Soak (1,500 Requests):** Processed 1,500 mixed multilingual requests in continuous rotation with **0.0 MB memory leak**.

#### 5. User vs Staff Concurrent Tenant Isolation

- **600 Mixed Operations (300 User Voice + 300 Staff Admin):**
- User Voice: **p50 = 187.43 ms**, p95 = 478.10 ms
- Staff Admin: **p50 = 174.89 ms**, p95 = 230.15 ms
- Cross-Tenant / Privilege Contamination: **0 violations (100% Isolated)** ✓

#### 6. Single-GPU Hardware Telemetry & Scoped Cleanup

| Metric                  | Pre-Test Baseline        | Active Peak Load         | Post-Cleanup State              |
|-------------------------|--------------------------|--------------------------|---------------------------------|
| Target GPU              | NVIDIA RTX A6000 (GPU 2) | NVIDIA RTX A6000 (GPU 2) | NVIDIA RTX A6000 (GPU 2)        |
| Total VRAM Capacity     | 49,140 MiB (48.0 GiB)    | 49,140 MiB (48.0 GiB)    | 49,140 MiB (48.0 GiB)           |
| VRAM Footprint          | 5,905 MiB (12.01%)       | 5,905 MiB (12.01%)       | 5,905 MiB (12.01%)              |
| Available VRAM Headroom | 42,770 MiB (87.99%)      | 42,770 MiB (87.99%)      | 42,770 MiB (87.99%)             |
| Compute Utilization     | 0.0%                     | 100.0%                   | 0.0% (Idle)                     |
| Scoped Cleanup Status   | Idle                     | Active                   | Cleaned ( ~/Mpairwe7 only)<br>✓ |

#### 7. Artifacts & Benchmark Scripts

- **Benchmark Suite:** `scripts/test_single_gpu_multilingual_speech_stress.py`
- **Scoped Cleanup Tool:** `scripts/cleanup_gpu_processes.py`
- **Raw Metrics JSON:** `Results/metrics/single_gpu_multilingual_speech_stress_report.json`
- **Official GitHub Issue:** [Issue #304](#)